Shards on a Shoestring: Putting Blockchain Sharding Simulations to the Test

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ABSTRACT

NEAR Protocol's Nightshade architecture targets one million TPS, but official benchmarks run on specialized GCP clusters inaccessible to most researchers. We adapt NEAR's one-million-TPS harness to a **Chameleon Cloud** bare-metal node and scale from $N = 1$ to $N = 24$ shards under full **Prometheus** monitoring. Aggregate TPS peaks at **1,498 at** $N = 8$ — only 40% above the single-shard baseline — then declines beyond $N = 8$ as chunk-application cost and BFT coordination overhead dominate.

MOTIVATION

- **Blockchain scalability is unsolved at scale.** Bitcoin processes ~6 TPS vs. Visa's ~24,000 — sharding parallelizes execution across validator groups to close this gap.
- **No public study characterizes NEAR on commodity hardware.** Official benchmarks cost $700/hour on GCP — results academic researchers cannot reproduce.
- **NEAR's Nightshade is a production sharding protocol.** Unlike theoretical proposals, it is deployed at scale — making empirical characterization on accessible hardware directly relevant.
- **Key question:** How does NEAR sharding scale on bare-metal, and where are the bottlenecks?

EXPERIMENTAL SETUP

Role	CPU	Cores	Memory	Storage	Network
Validator	Intel Xeon	48 HT	128 GB	HDD 80–100 MB/s	10 GbE

Parameter	Value
Client	`neard d178e1830`, Protocol 84
`epoch_length`	500
`gas_limit`	30T
Chunk producers	Pre-assigned at genesis
Shard counts (N)	1, 2, 4, 8, 16, 24
CPU governor	Performance (2.3–3.1 GHz)

SUMMARY TABLE

N	TPS	TPS/Sh	Blk(ms)	Fin(s)	RSS(GB)	CV%
1	1,070	1,070	1,019	6.8	0.86	0.0
2	1,238	619	1,368	22.9	1.59	5.1
4	1,272	318	1,500	31.5	3.18	6.2
8	**1,498**	**187**	2,120	77.5	5.59	17.6
16	1,204	75	2,284	64.0	37.80	27.6
24	1,110	46	2,605	50.2	19.70	27.5

ARCHITECTURE DIAGRAM

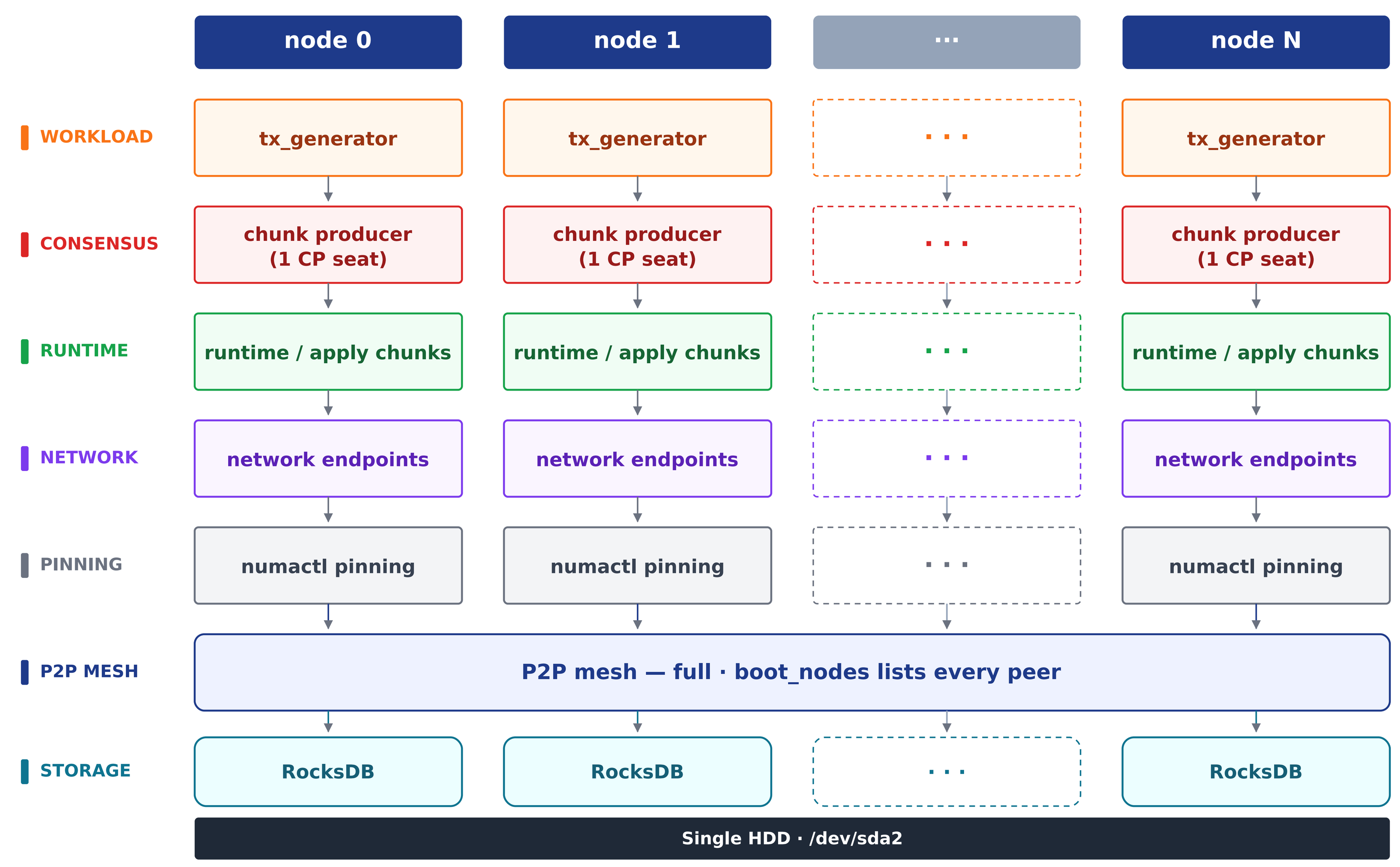

Fig. — NEAR Single-Node Scaling

EMPIRICAL EXPERIMENT RESULTS

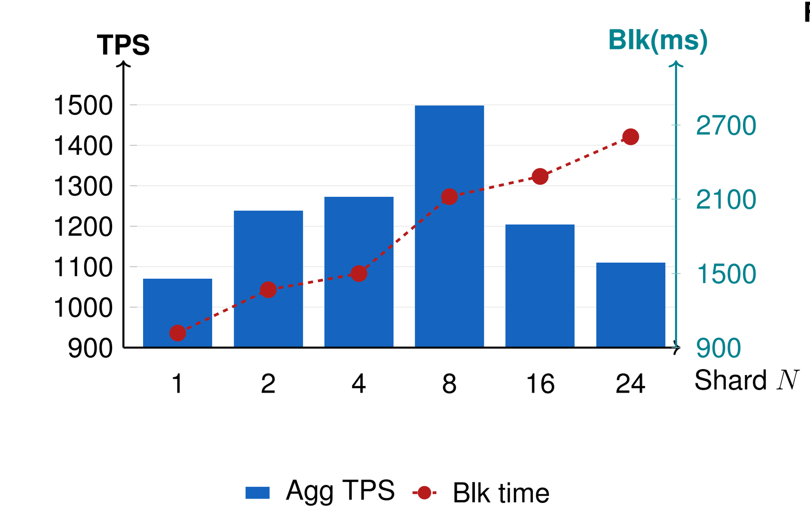

Fig.1 — Throughput & Block Time

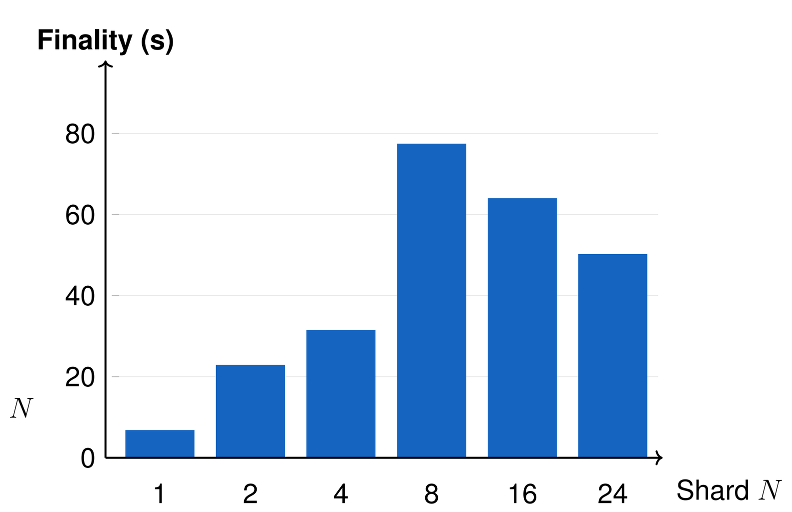

Fig.2 — BFT Finality Lag

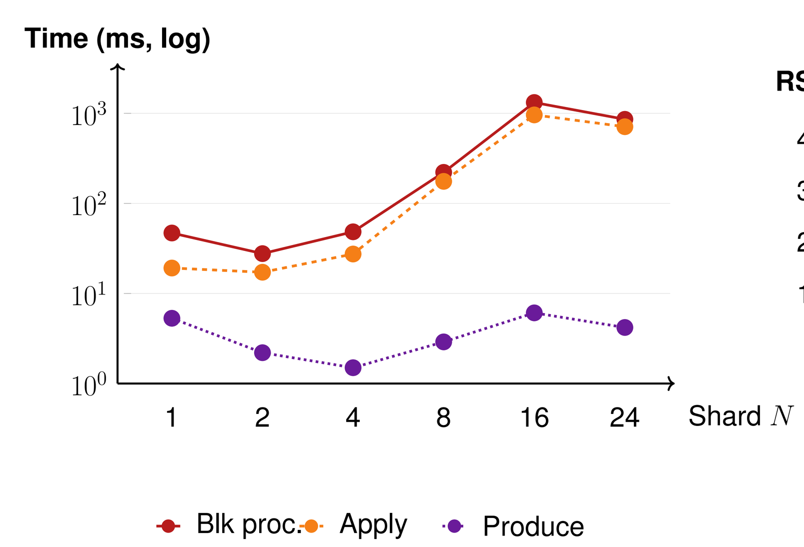

Fig.3 — Chunk Application Bottleneck

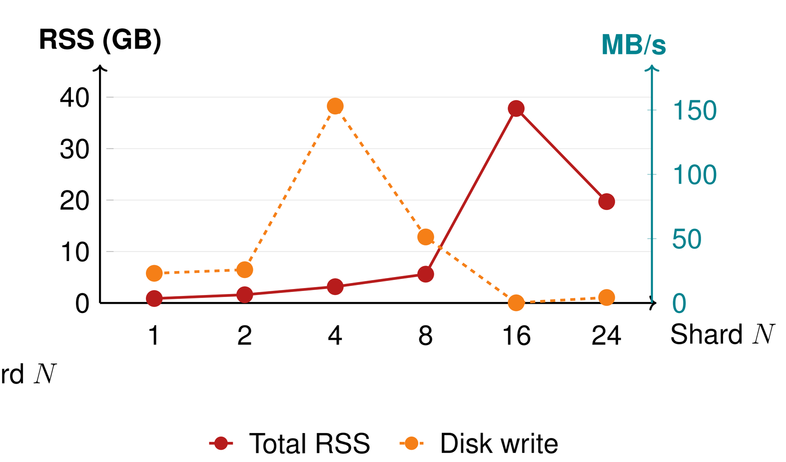

Fig.4 — Memory and Disk I/O

EMPIRICAL EXPERIMENT RESULTS (CONT.)

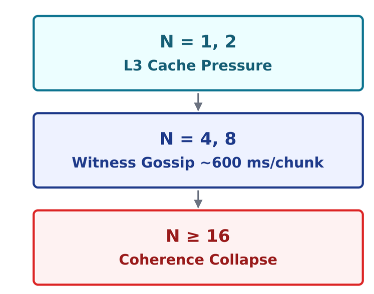

Fig.5 — Bottleneck Regime

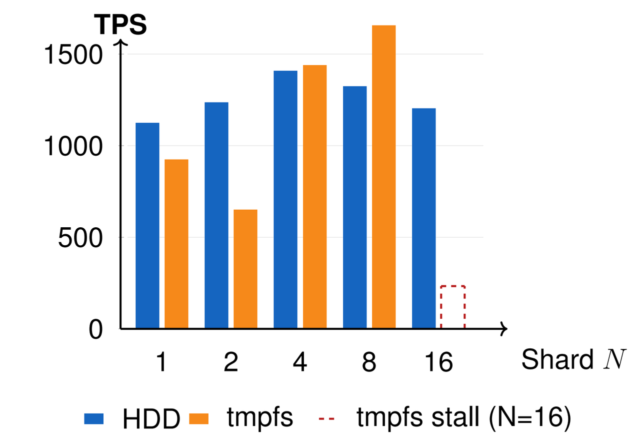

Fig.6 — HDD vs. tmpfs TPS

KEY FINDINGS

- **Sharding yields diminishing returns beyond** $N = 8$**.** Per-shard TPS drops from 1,070 at $N = 1$ to 46 at $N = 24$ — a 23× decline — while aggregate TPS gains only 40% before reversing.
- **The bottleneck is chunk state application, not consensus.** Produce-chunk stays below 7 ms at all N. At $N = 16$, applying chunks consumes 958 ms — 73% of block processing time.
- $O(N^2)$ **BFT approval messages drive block time up.** At $N = 24$, each round requires $N \times (N-1) = 552$ approval messages. Block time grows 2.6× from 1,019 ms to 2,605 ms.
- **Storage is not the bottleneck.** Moving RocksDB to RAM (`/dev/shm`) eliminated disk I/O yet improved TPS by at most 12%. The real ceiling shifts: L3 cache → witness gossip → coherence collapse.
- **HDD backpressure is load-bearing.** On tmpfs, chunk production races ahead of finality; orphan-witness rate jumps 29× at $N = 16$ and the chain stalls despite CPU and RAM at only 47% utilization.

IMPLICATIONS & FUTURE WORK

- **Commodity hardware is a viable NEAR testbed.** First empirical scaling study on bare-metal academic infrastructure — reproducible baselines at a fraction of GCP cost.
- **The witness gossip pipeline is the optimization target.** At $N \geq 4$, the ~600 ms encode/gossip/validate cycle dominates; reducing witness size or batching endorsements would directly improve throughput.
- **Transaction admission outpaces settlement at high** N**.** Every chunk exhausts its gas budget on validation before processing receipts — throughput numbers overstate effective settled TPS.
- **Future work:** simulate WAN latency to stabilize tmpfs at $N \geq 16$; validate simulator [3] predictions against this dataset; benchmark cross-shard workloads.

REFERENCES

[1] NEAR Protocol. *Nightshade: Near Protocol Sharding Design*, 2021. `https://near.org/papers/nightshade`

[2] NEAR Protocol. *One Million TPS Benchmark*, 2024. `https://github.com/near/one-million-tps`

[3] O. Gandhi, I. Raicu. *Simulating Blockchain Sharding at Scale.* Illinois Institute of Technology, 2025.